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High temperature oxidation behaviors of equimolar NbTiZrV and NbTiZrCr refractory complex concentrated alloys (RCCAs)

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ABSTRACT

Refractory complex concentrated alloys (RCCAs) represent an emerging class of materials that have displayed significant potential for increased mechanical performance at extreme temperatures over conventional refractory alloys and Ni-base superalloys. While limited work has been conducted in this area, several RCCAs have shown remarkable levels of oxidation resistance in comparison to the majority of uncoated commercial refractory alloys. However, the oxidation mechanisms in these types of alloys have not been well understood. This study aims to develop the foundation for understanding RCCA oxidation mechanisms through the systematic evaluation of NbTiZrV and NbTiZrCr, which display vastly different oxidation behaviors. In both cases, the predominant mechanism appeared to be internal oxidation, where a mixture of simple and complex oxides was observed after prolonged oxidation times. This resulted in extended linear oxidation kinetics for the NbTiZrV alloy and linear to near-parabolic oxidation kinetics for the NbTiZrCr alloy. Under this mindset, novel design strategies for promoting increased oxidation resistance in both RCCAs and dilute refractory alloys based on sluggish complex oxides are proposed.

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1. Introduction

Refractory complex concentrated alloys (RCCAs), which include refractory high-entropy alloys (RHEAs), are an innovative class of materials that are based on three or more refractory metals (e.g. Cr, Hf, Nb, Mo, Ta, V, W, Zr) and commonly other elements, such as Al, Si, and Ti in high concentrations [1–3]. This design strategy utilizes high-melting point refractory elements to promote high temperature strength, sufficient quantities of light elements (e.g. Al, Ti, Zr) to promote reduced alloy densities, and elements like Al, Cr, and/or Si to promote oxidation resistance through the formation of passivating oxide scales. Starting from the pioneering work by Senkov et al. [4,5] on NbMoTaW and NbMoTaVW alloys, which showed yield strength above 400 MPa at temperatures up to 1600 °C, these complex configurations have been reported to facilitate promising mechanical performance with reduced densities, with some alloys surpassing the temperature capabilities of Ni-base superalloys used for blade and disk turbine engine applications [5–9]. For example, Han et al. [7] investigated the role of Ti additions on the phases and mechanical response of NbMoTaW and NbMoTaVW based alloys up to 1200 °C. It was ultimately determined that Ti exhibited the potential to reduce density, improve high temperature strength, and increase compressive ductility at room temperature. Senkov et al. [8] reported several Al-containing RCCAs with reduced density and superior strength both at room and elevated temperatures. A characteristic microstructure typical to superalloys and consisting of cuboidal BCC nanoscale precipitates orthogonally-aligned within a coherent B2 matrix was identified to be responsible for the high strength of these alloys [9,10]. These efforts triggered other recent works exploring comparable RCCAs [11–14]. Alternatively, Guo et al. [15], Zhang et al. [16] and Liu et al. [17] explored the role of Si in promoting high strength silicide phases in $Mo_{0.5}NbHf_{0.5}ZrTi/M_5Si_3$, $HfNbTiVSi_{0.5}$ and $HfMo_{0.5}NbTiV_{0.5}Si_x$ RCCAs. In all the cases, moderate alloying additions of Si promoted the formation of M_5Si_3 type phases (where M = Nb,Hf,Ti), which exhibited the potential to improve high temperature strength (up to 1200 °C) and increase room temperature ductility. Nevertheless, these types of composites are still susceptible to strain softening and silicide segmentation during high-temperature mechanical testing.

In comparison to the abundance of work emphasizing phase

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equilibria and mechanical phenomena, only a limited number of papers have investigated the oxidation behaviors of RCCAs relative to commercial refractory alloys [18–21]. Poor environmental resistance is a large limitation of transitioning commercial refractory alloys and requires the use of prime reliant EBC coatings [22]. In comparison to the majority of uncoated commercial refractory alloys (e.g. C-103, C-129Y, Cb-752, TZM), which rapidly oxidize at high temperatures, preliminary oxidation results on uncoated RCCAs display surprising levels of oxidation resistance [18–21]. However, the oxidation mechanisms responsible for the improved oxidation resistance of these alloys are not well understood. This understanding is crucial for the design and development of RCCAs for potential use in high temperature, oxidizing environments.

The majority of published work on the oxidation behaviors of RCCAs tends to focus on promoting the formation of continuous protective oxide scales (e.g. Al_2O_3, Cr_2O_3, and/or SiO_2) through alloying additions of Al, Cr, and/or Si [19–21]. However, there is also a small subset of papers that have explored the oxidation behaviors of mechanically optimized RCCAs [18,21,23]. Gorr et al. [19,21,23] investigated the oxidation behavior of several Mo-Cr-Ti-Al-X RCCAs, where X = Nb, W, or Ta. Their design strategy was to combine high melting point elements (e.g. Mo, Nb, W) with high concentrations of stable oxide formers (e.g. Al and/or Cr) and low density elements like Ti, to facilitate high-strength, low-density refractory alloys with some level of oxidation resistance. It was reported that the Mo-Cr-Ti-Al-W and Mo-Cr-Ti-Al-Nb alloys did not obey parabolic oxidation kinetics at 1000 °C and 1100 °C out to 40 h. However, the magnitude of specific mass change in both cases was considerably low (~0.25 mg/cm^2 per hour) compared to those commonly observed for pure refractory metals (>20–60 mg/cm^2 per hour [24–26]) and their alloy containing Nb exhibited the smallest mass change. It was deduced that an increased oxidation rate of the W-containing alloy was due to the unfavorable reaction of Al_2O_3 with WO_3 to form a non-protective $Al_2(WO_4)_3$ oxide that inherently reduced the probability of forming a continuous Al_2O_3 layer. In contrast, the Mo-Cr-Ti-Al-Ta alloy exhibited specific mass changes on the order of Cr_2O_3-formers and was reported to adhere to parabolic oxidation kinetics out to 40 h at both 1000 °C and 1100 °C. The superior oxidation kinetics of this alloy was reported to be due to the formation of a continuous Al_2O_3 scale that formed under an external TiO_2 scale. In addition, a $CrTaO_4$ phase was identified, but its influence on oxidation resistance was not addressed. In all three cases, the volatility of Mo was reported to be insignificant. Gorr et al. [21] also examined the influence of small additions of Si on the resulting oxidation behavior of Mo-Cr-Ti-Al-Nb at 900 °C, 1000 °C, and 1100 °C. The presence of Al and Cr were reported to promote some extent of Al and Cr rich layers (reported to be Al_2O_3/Cr_2O_3), but were accompanied by other native and higher order oxides (e.g. TiO_2, $Al_2(WO_4)_3$). It was determined that Si possessed a beneficial influence by promoting parabolic oxidation kinetics, but no clear mechanism was proposed. These results are also consistent with the work of several other groups [18,20,27] who reported highly complicated mixtures of higher order oxides (e.g. $CrVNbO_6$, $NbCrO_4$) with Cr and/or Al containing RCCAs, but observed significantly higher mass increases. Nevertheless, these latter cases still demonstrate a substantial improvement in oxidation resistance compared to uncoated commercial refractory alloys, without necessarily relying on the formation of continuous/protective Al_2O_3, Cr_2O_3, and/or SiO_2 scales. Their kinetically sluggish, complex oxides could have significant potential for use in short-term, expendable applications in highly aggressive environments and should also be considered as an alternative refractory alloy design approach.

The intent of this work is to establish the foundation for understanding the complex oxidation behaviors of RCCAs. Unlike the majority of the literature, the goal is not necessarily to only facilitate the formation of common protective oxides (Cr_2O_3, Al_2O_3, SiO_2); rather, it is to fundamentally address the influence of chemistry on the underlying oxidation mechanisms and oxidation kinetics in these complicated systems. This includes understanding the various degrees of oxidation resistance provided by novel, complex oxides as well. In particular, NbTiZrCr and NbTiZrV were investigated. The comparison of these two alloys provides insight on the actual influence of alloying with Cr, which is assumed to provide enhanced oxidation resistance due to the formation of Cr_2O_3. V and Cr are also neighboring elements in the Periodic Table, and have the same BCC crystal structure, almost the same melting temperature, similar oxidation states and similar electronegativity. However, the effects of these elements on the kinetics of oxidation and oxidation resistance of the NbTiZr-X alloys are found to be remarkably different. The results presented here will aid in the development of a design methodology relevant for the optimization of RCCAs for high temperature applications.

2. Experimental procedures

Two bulk alloys with nominal compositions (in at.%) of $Nb_{25}Ti_{25}Zr_{25}V_{25}$ and $Nb_{25}Ti_{25}Zr_{25}Cr_{25}$ were vacuum arc melted from high purity constituents on a water-cooled copper hearth. To mitigate contamination from residual oxygen and nitrogen, high purity Ti was used as an initial getter prior to primary melting. The alloys were flipped and re-melted at least five times to ensure homogeneity and eventually cast into cigar-shaped ingots measuring approximately 100 mm long × 30 mm wide × 12 mm tall. The experimental bulk alloy compositions were found to be in good agreement with the target compositions, as determined by inductively coupled plasma-optical emission spectroscopy (ICP-OES), Table 1. Both alloys were then wrapped in Ta foil and hot isostatically pressed (HIPed) at 1400 °C (NbTiZrV) / 1200 °C (NbTiZrCr) for 2 h and 207 MPa. This was followed by a 1200 °C anneal for 24 h under an inert, high-purity argon atmosphere with a subsequent cooling at the rate of 10 °C/min.

Specimens for microstructural analysis and oxidation testing were excised from the central regions of each ingot via electric discharge machining (EDM). Continuous oxidation tests of the (HIPed + homogenized) alloys were conducted at 1000 °C in dry air using a Thermax 700 Thermogravimetric Analyzer (TGA). The temperature of 1000 °C was selected because RCCAs are expected to operate at this or higher temperatures. Another reason for selecting this temperature is that it has often been used during oxidation experiments with other RCCAs and concurrent alloys, thus allowing comparison. Prior to exposure, test coupons were ground to a 600 grit finish and ultrasonically cleaned using acetone followed by isopropanol. The dimensions of the prepared samples were measuring ~4.5 mm × 4.5 mm × 7.0 mm. The TGA furnace was

Table 1
The overall alloy compositions as determined by inductively coupled plasma optical emission microscopy (ICP-OES) and the respective phase compositions as determined via SEM-EDS. All data is represented in at. %.

Alloy	Region	Cr	Nb	Ti	V	Zr
NbTiZrV	Overall (ICP-OES)	–	28.3	24.5	23.0	24.2
	BCC **(Matrix)**	–	27.7	24.4	23.7	24.2
	BCC_2 **(Dark)**	–	22.1	20.5	38.9	18.5
	White Near Dark	–	24.4	22.1	21.0	32.5
NbTiZrCr	Overall (ICP-OES)	24.6	26.7	23.9	–	24.8
	BCC **(Bright)**	7.9	34.6	31.4	–	26.1
	Laves C15 **(Dark)**	51.5	15.8	12.1	–	20.6

heated under continuously flown dry air (80 ml/min) at 15 °C/min up to 1000 °C, held at 1000 °C for 3.60×10^5 s (100 h), and cooled at a rate of ~10 °C/min. Only mass data collected at the 1000 °C set-point was considered in this work. In order to better understand initial oxide scale formation, abbreviated oxidation tests were also conducted in a box furnace under static lab air for 1800 s (0.5 h) and 1.44×10^4 s (4 h).

Prior to microstructural characterization, non-oxidized and oxidized specimens were mounted in phenolic resin and polished using standard metallographic techniques. Final polishing was conducted using 0.05 μm colloidal silica on a vibratory polisher. The alloy phase structures, chemistries, and morphologies were investigated using scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), transmission electron microscopy (TEM), and X-ray diffraction (XRD). Plan-view and cross-sectional images of the specimens were taken on an FEI Quanta 600F SEM in both secondary electron (SE) and backscattered electron (BSE) modes at 20 kV. TEM foils were made using a variation of the focused-ion-beam (FIB) in situ lift-out technique using an FEI Nova 600 Dual Beam FIB-SEM [28]. High angle annular dark field (STEM-HAADF) images and selected area electron diffraction patterns (SADPs) were captured on a 200-KeV FEI Talos F200X (S)TEM equipped with a Super-X (Quad) EDS system. XRD spectra were collected in plan-view from bulk specimens using a Bruker D2 PHASER XRD with Cu-Kα radiation and a 2θ scan range of 20–80°.

3. Results

3.1. HIPed + annealed alloy microstructures

The microstructures of the NbTiZrV and NbTiZrCr alloys have been investigated in great detail elsewhere [6,29], but will be briefly summarized here to provide some degree of context for the resulting oxidation behaviors. Fig. 1 shows representative BSE images from each HIPed + annealed alloy. The NbTiZrV alloy was extremely coarse grained and exhibited a two-phase microstructure consisting of a BCC matrix phase with sub-micron BCC precipitates, Fig. 1(a)–(b). These V-rich precipitates were found to be spatially arranged in chain-like morphologies and were surrounded by high-Z contrast regions. The respective phase compositions (at. %) based on SEM-EDS are shown in Table 1. Fig. 1(c)–(d) show representative BSE images of the NbTiZrCr alloy. The microstructure was comprised of a BCC phase, enriched in Nb, Ti, and Zr, and a Laves phase that was enriched in Cr (Table 1).

3.2. Oxidation behaviors

Fig. 2(a) shows specific mass gain (normalized by initial surface area) vs oxidation time for the NbTiZrV (black solid line) and NbTiZrCr (red dashed line) alloys. The same curves are plotted on a Log/Log scale in Fig. 2(b). It is clear from these curves that the V-containing alloy displayed considerably faster oxidation rates, which are more comparable to pure refractory metals and their alloys [30,31]. After approximately 2.9×10^4 s (8 h) of oxidation, the NbTiZrV alloy was completely oxidized, as indicated by the plateau in mass change (labeled "All Oxide") in Fig. 2(b) and is consistent with the inset digital image in Fig. 2(a) taken after 3.6×10^5 (100 h) of oxidation. In contrast, the NbTiZrCr alloy displayed much lower mass gains and was comprised of oxide and remaining "bulk" after 3.6×10^5 s (100 h) of oxidation exposure, as shown by the inset digital image in Fig. 2(a). The mass curves in Fig. 2(a) were fitted to a general oxidation growth law with [31]:

$$\Delta m = kt^n$$

where Δm is specific mass change, k is the oxide growth rate

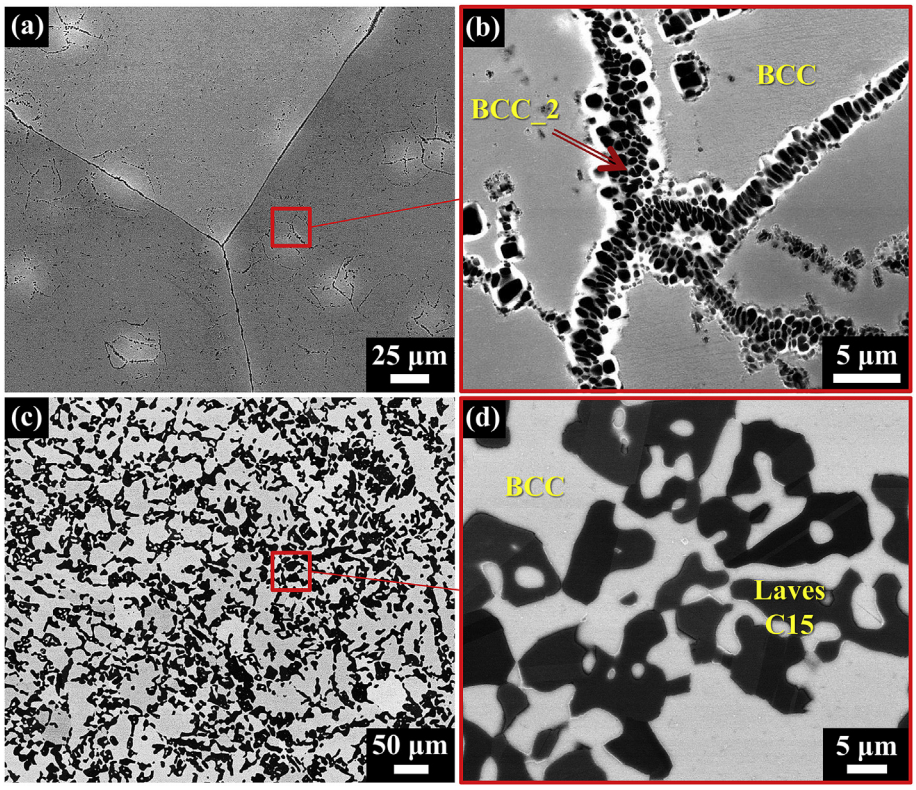


Fig. 1. BSE images from the NbTiZrV (a)–(b) and NbTiZrCr (c)–(d) alloys in the HIPed + annealed state. The phases are labeled based on their respective structures.

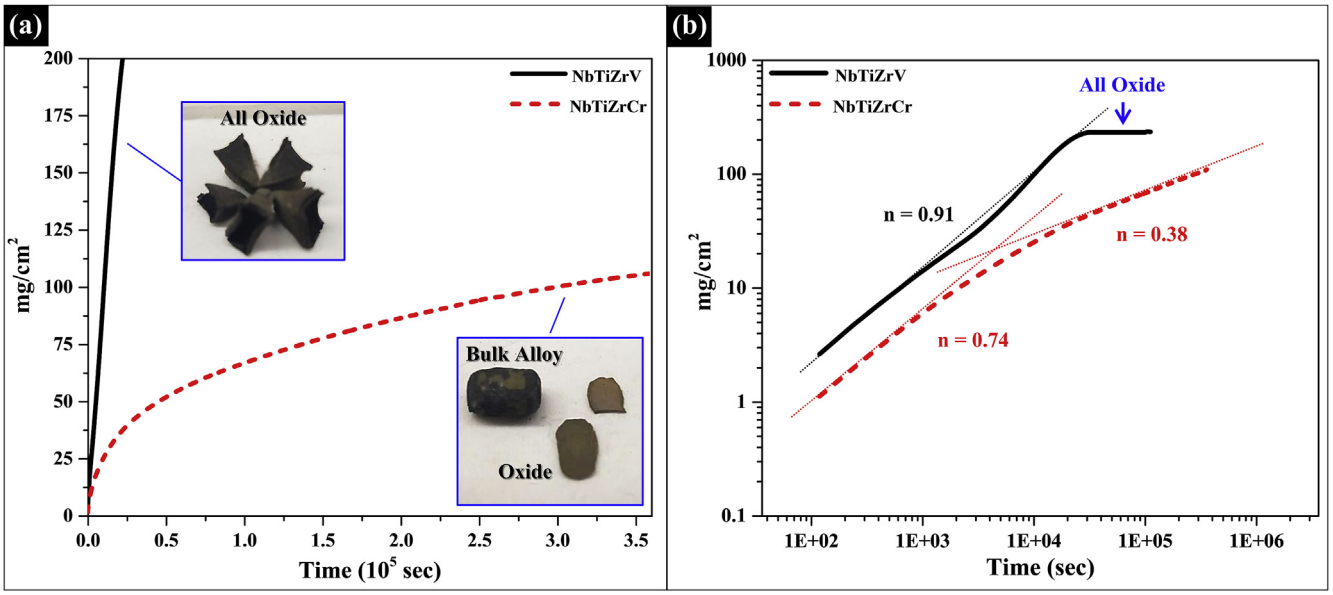


Fig. 2. Specific mass change plots for the NbTiZrV and NbTiZrCr alloys subjected to oxidation at 1000 °C on a linear scale (a) and on Log/Log scales (b). Inset digital images in (a) show the respective alloys after 100 h of oxidation. Growth rate constants "n" in (b) represent linear fits for each respective curve and region.

constant, t is time, and n is the time exponent. The best fit for the NbTiZrV alloy was found to be linear ($n = 1$) with $k = 9.4 \times 10^{-3}$ mg cm^{-2} s^{1}. The best fit for the NbTiZrCr alloy was non-linear and closer to parabolic and/or cubic behavior ($n = 0.38$) with $k = 6.6 \times 10^{-4}$ mg cm^{-2} s$^{-0.38}$. These behaviors were also investigated on Log/Log coordinates, as shown in Fig. 2(b). The NbTiZrV alloy maintains a rapid, approximately linear rate of oxidation ($n = 0.91$) throughout the entire duration of testing. A small dip in the curve was observed from ~757 s (0.21 h) to ~1.2×10^4 s (3.2 h) in the initial portion of the test. This is likely due to oxide partial separation and/or cracking (e.g. along the edges), which exposes new metal free surface to oxidation that drives a slight decrease followed by secondary acceleration. For the NbTiZrCr alloy, the initial oxidation rate was fairly rapid ($n = 0.74$) from ~0 to 6581 s (0–1.8 h) and then transitions to considerably slower kinetics ($n = 0.38$). This suggests that the NbTiZrCr alloy undergoes some early change in oxidation mechanism, which is then sustained for the remainder of testing. However, in all cases, the deviation from ideal linear ($n = 1$) and parabolic ($n = 0.5$) behaviors, as shown by the Log/Log plots, indicates the activation of multiple mechanisms during oxidation testing.

3.3. Microstructure analysis

3.3.1. Long term oxidation

Fig. 3 shows representative BSE images of the oxides after 3.6×10^5 s (100 h) of oxidation. The NbTiZrV oxide scale was composed of a several different oxides, as shown by the various morphologies found at the free surface in the plan-view image in Fig. 3(a). Some of the oxides formed continuous and fairly dense layers, while others exhibited a rod-like character and were dispersed all over the surface plane. Cross-sectional imaging revealed a fairly thick portion of fully oxidized metal (~3235 μm) with an outward expanding geometry, likely due to the volumetric change accompanied by complete oxide formation of the base metal, Fig. 3(b). As a consequence, the free surface was also observed to be heavily curved. The sample showed some evidence of initial, external oxide scale formation during the early stages of oxidation, as evident from the high-Z contrast region labeled with arrows near the free surface, Fig. 3(b). Higher magnification imaging near the center of the NbTiZrV scale revealed a porous and complex distribution of three discernable oxides, Fig. 3(c). As shown in Fig. 4(a) and Table 2, the three oxides consisted of high-Z contrast needles rich in Zr/Nb (labeled 1), larger medium-Z contrast islands rich in Nb (labeled 2), and small low-Z contrast regions rich in Ti (labeled 3). The chemical analysis revealed a considerably reduced amount of V in the oxide scale, relative to the V concentration in the alloy before oxidation. This observation indicates that V was likely to form a volatile oxide, which evaporated during 100 h holding at 1000 °C or was removed during sample preparation by reaction with the polishing solution/media, as it is soluble in water. Since the entire NbTiZrV bulk specimen completely oxidized, as was shown in Fig. 2(a), no analysis of remaining "bulk" material could be conducted.

The 3.6×10^5 s (100 h) oxide scales from the NbTiZrCr alloy were much more homogeneous in comparison to the NbTiZrV scales. Fig. 3(d) shows a plan-view BSE image with little variation in compositional contrast throughout. Cross-sectional imaging revealed a much thinner scale (~467 μm) with no dramatic curvature at the free surface, Fig. 3(e). In contrast to the NbTiZrV alloy, the free surface did not display evidence of significant external oxide scale formation during the initial stages of oxidation (e.g. no distinct boundaries/oxides at the free surface of the oxide scale). The oxide scale also exhibited micro-scale compositional contrast variations distributed through the entire thickness that resembled the former BCC/Laves phase distribution, Figs. 1(c)–(d) and 3(e)–(f). However, as shown in Fig. 4(b) and Table 2, the relative compositions of each region were fairly similar and contained nearly equal amounts of all constituents. The only significant difference between the two appeared to be the Cr content, which was higher in the darker regions (labeled 4) and lower in the lighter lamellar regions (labeled 5), Fig. 4(b). It is also important to note that the relative change in the Z-contrast on the BSE image could also be due to a phase orientation effect with respect to the viewing plane or the presence of an additional phase (light lamellae or chains of fine particles) with a reduced concentration of Cr.

The NbTiZrCr alloy did have remaining "bulk" material after 3.6×10^5 s (100 h) of oxidation testing, as shown by the cross-

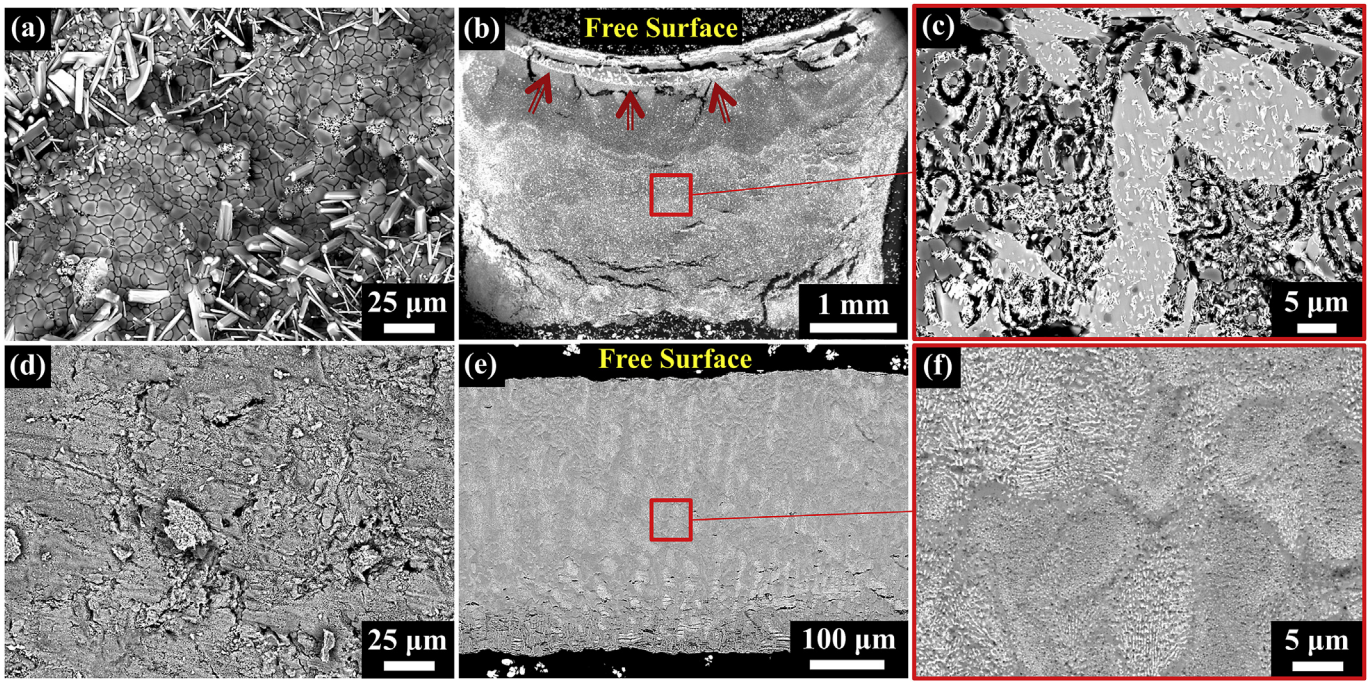


Fig. 3. Representative plan-view, cross-section, and higher magnification images of the 100hr oxides for the NbTiZrV alloy (a)–(c) and NbTiZrCr alloy (d)–(f). The inset boxes in (b) and (e) represent the relative locations of the higher magnification images in (c) and (f). The location of the outer most surface of the oxide is labeled "free surface".

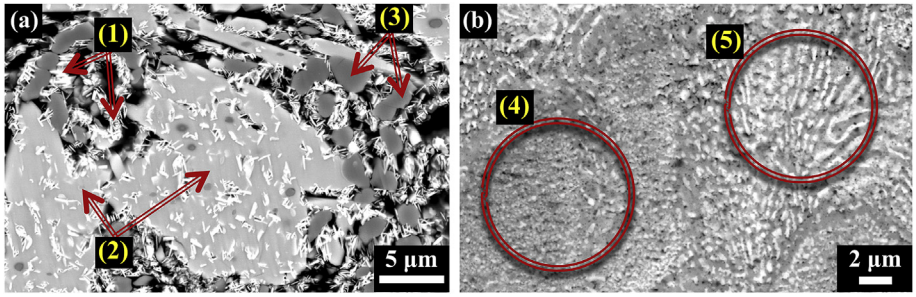


Fig. 4. BSE images from the center of the 100hr oxides for the NbTiZrV alloy (a) and NbTiZrCr alloy (b). The respective locations for each SEM-EDS measurement show in Table 2 is labeled via number in (a) and (b).

Table 2
Phase compositions from the 100hr oxide scales formed on the NbTiZrV and NbTiZrCr alloy as determined via SEM-EDS. All data is represented in at. %. The respective locations are identified in Fig. 4.

Alloy	Region	O	Cr	Nb	Ti	V	Zr
NbTiZrV	White Needles **(1)**	69.0 ± 0.2	–	11.1 ± 0.7	2.7 ± 0.4	0.9 ± 0.1	16.3 ± 1.3
	Large Gray **(2)**	69.4 ± 0.4	–	17.7 ± 0.3	5.0 ± 0.2	2.9 ± 0.1	5.1 ± 0.5
	Dark Gray **(3)**	71.9 ± 1.1	–	4.5 ± 0.4	15.3 ± 0.6	2.3 ± 0.2	6.0 ± 1.2
NbTiZrCr	Overall Dark **(4)**	61.0 ± 0.7	11.1 ± 2.9	10.1 ± 1.9	9.0 ± 0.5	–	8.9 ± 1.3
	Overall Brighter **(5)**	60.9 ± 0.1	4.9 ± 0.6	13.8 ± 0.4	10.6 ± 0.2	–	9.7 ± 0.5

sectional BSE images in Fig. 5. A significant portion of un-oxidized "bulk" remained, but internal oxidation was observed to penetrate ~750 μm into the surface (in reference to the interface where the oxide fractured/separated from the main specimen), Fig. 5(a). During internal oxidation, the BCC matrix formed a eutectoid-like, lamellar mixture, while the prior Laves phase was more uniform in Z-contrast and displayed a high-Z phase along boundaries and contained dark particles internally, Figs. 5(b) and 6(b). As expected, the extent of oxidation decreased with penetration depth, as indicated by the variation in Z-contrast in Fig. 5(a). In addition, a series of abrupt internal transitions were observed at various depths bounding similar microstructures and are likely due to differences in local partial pressure of oxygen, Figs. 5(c) and 6. The extent of internal oxidation is also consistent with the relative region compositions, as determined via EDS, shown in Fig. 6 and Table 3 (regions 1, 2, 3). The oxygen content was found to decrease in each region toward the un-oxidized central core. Furthermore, the internally oxidized Laves phase was extensively rich in Cr (labeled 4), while the lamellar BCC region was richer in oxygen and lean in Cr (labeled 5).

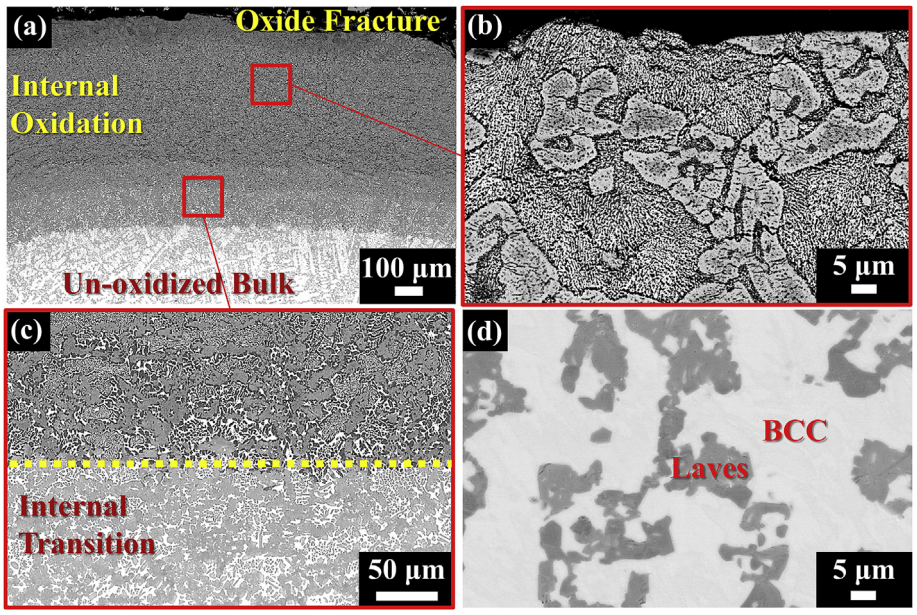


Fig. 5. Representative cross-section BSE images captured from the remaining bulk portion of the NbTiZrCr alloy after 100 h of oxidation at 1000 °C. (a)–(c) show regions near the internal oxidation, while the central "un-oxidized" core of the alloys is shown in (d).

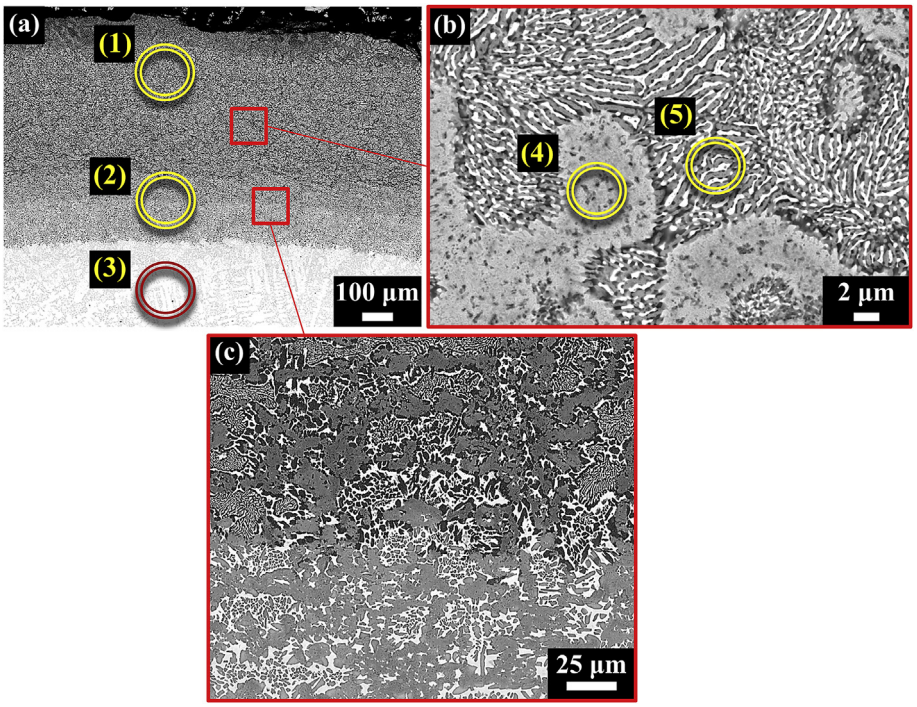


Fig. 6. Representative cross-section BSE images captured from the remaining bulk of the NbTiZrCr alloy after 100 h of oxidation at 1000 °C (a)–(c). The respective locations for each SEM-EDS measurement shown in Table 3 is labeled via number in (a) and (b).

To gain a better understanding of the oxide products, TEM analysis was conducted on specimens lifted out near the center of the 3.60×10^5 s (100 h) oxide scales, as shown by the relative location of the inset boxes in Fig. 3(b) and (e), respectively. The lift-outs were strategically done across distinct phase boundaries in order to capture all discernable oxides in the long-term oxide scales. Fig. 7 shows STEM-HAADF images from the NbTiZrV oxide, along with high-resolution elemental chemistry maps. Based on Z contrast alone, the three distinct regions observed via SEM were also present in the STEM-HAADF images, Figs. 4(a) and 7(a). In addition, the elemental maps revealed strong partitioning of Zr (green) to the needle-like phase, Ti (blue) to the round/globular phase, and Nb (red) to the large plate-like phase, Fig. 7(b)–(e). In contrast, V (cyan) tended to prefer multiple phases and enriched the bottom side of the TEM foil, Fig. 7(a) and (c). This could suggest that SEM sample prep influenced the amount of V and/or V-rich oxides near the polished surface due to reaction with polishing solution/media. This is also supported by the idea that "trenching"

Table 3
Overall compositions taken from internally oxidized regions of the NbTiZrCr alloy after 100 h of oxidation, as determined via SEM-EDS. All data is represented in at. %. The respective locations are identified in Fig. 6.

Region	O	Cr	Nb	Ti	Zr
Top **(1)**	45.3 ± 0.3	12.1 ± 1.4	15.2 ± 0.4	13.5 ± 0.3	13.9 ± 0.2
Mid **(2)**	32.1 ± 1.2	17.8 ± 0.7	17.1 ± 0.2	17.5 ± 0.4	15.5 ± 0.3
Bulk **(3)**	7.8 ± 2.0	22.8 ± 1.2	23.4 ± 0.3	22.3 ± 0.2	23.7 ± 0.5
Island **(4)**	37.5 ± 0.6	29.4 ± 0.8	11.1 ± 0.5	9.0 ± 0.6	13.0 ± 0.3
Lamellar **(5)**	48.1 ± 0.3	2.2 ± 0.1	21.3 ± 0.4	14.3 ± 0.2	14.1 ± 0.7

during TEM foil prep was conducted in cross-section, which leaves the bottom region of the foil relatively unaltered from grinding/polishing and seemingly enriched in V content, Fig. 7(c). Several porous regions of the TEM foil were infiltrated by mounting epoxy during sample prep, as shown by the STEM-HAADF image and corresponding C elemental map in Fig. 7(f)–(g).

In order to visualize the various oxides present, the elemental chemistry maps were stacked (including Gallium) into a composite map, as shown in Fig. 8(a). Oxide structures were determined using selected area electron diffraction (SAED) tilting experiments on each distinct oxide phase. Several zone axes (SADPs) were captured for each phase and correlated to an existing prototype structure with similar chemistry. Fig. 8(b)–(d) show indexed, representative SADPs for the three distinct oxides. The red, Nb-rich plate-like phase (b) exhibited a structure consistent with $TiNb_2O_7$ (ICDD 00-039-1407, I2/m (No. 12)). The blue, Ti-rich globular phase (c) matched TiO_2 (ICDD 00-021-1276, P42/mnm (No. 136)). The green, Zr-rich phase (d) exhibited a structure analogous to $Nb_2Zr_6O_{17}$ (Ima2 (No. 46)), which can be synthesized by reaction of Nb_2O_5 with ZrO_2 [32]. In all three cases, the respective lattice parameters were not found to be significantly different compared to the prototype structures. This is not too surprising since Nb, Ti, Zr, V all share similar atomic radii. No distinguishable V-oxide structures could be determined via SAED in the cyan regions of Fig. 8(a). However, based on the chemical maps, it is highly likely that some V-oxides (or V enrichment) are present to some degree.

The proposed oxide structures for the NbTiZrV alloy were further compared to the experimentally determined oxide compositions (TEM-EDS), as shown in Table 4. The compositions are shown without oxygen to compare the relative ratios of metallic constituents for each of the regions. The affinity for elements to preferentially substitute on the same lattice sites was based on their location in the Periodic table (i.e. elements from the same column such as Ti/Zr were assumed to prefer the same lattice sites in the respective oxides). The blue regions, matching a TiO_2 structure, were extensively rich in Ti and identified as TiO_2 with some residual background phases. There is likely some solubility of the other constituents in TiO_2, but this could not be trivially determined based on the chemical results in this study. Similarly, the green regions, which exhibited a $Nb_2Zr_6O_{17}$ structure, chemically matched $(Nb+V)_2(Zr+Ti)_6O_{17}$ when considering the ratios of Nb/V and Zr/Ti. The green and cyan regions likely consist of the same $(Nb+V)_2(Zr+Ti)_6O_{17}$ phase with some additional amount of V-oxide. Lastly, the red regions, identified with a $TiNb_2O_7$ structure, had a chemistry fairly consistent with $(Ti+Zr)(Nb+V)_2O_7$. Thus, the

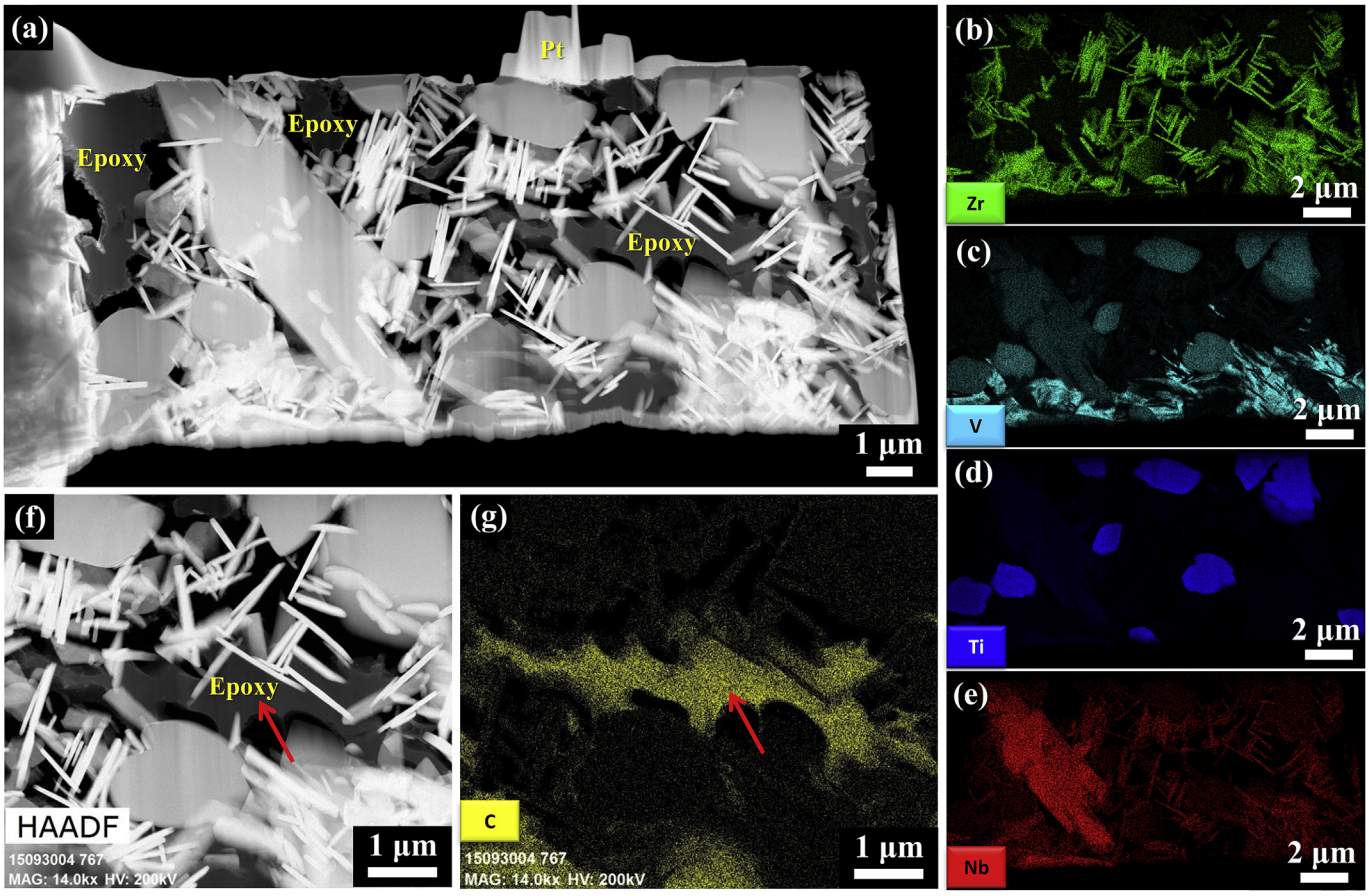


Fig. 7. STEM-HAADF image of a specimen lifted-out of the center of the NbTiZrV 100hr oxide scale (a) and corresponding elemental TEM-EDS maps (b)–(e). An additional higher magnification STEM-HAADF image taken from (a) and corresponding carbon TEM-EDS map are shown in (f)–(g).

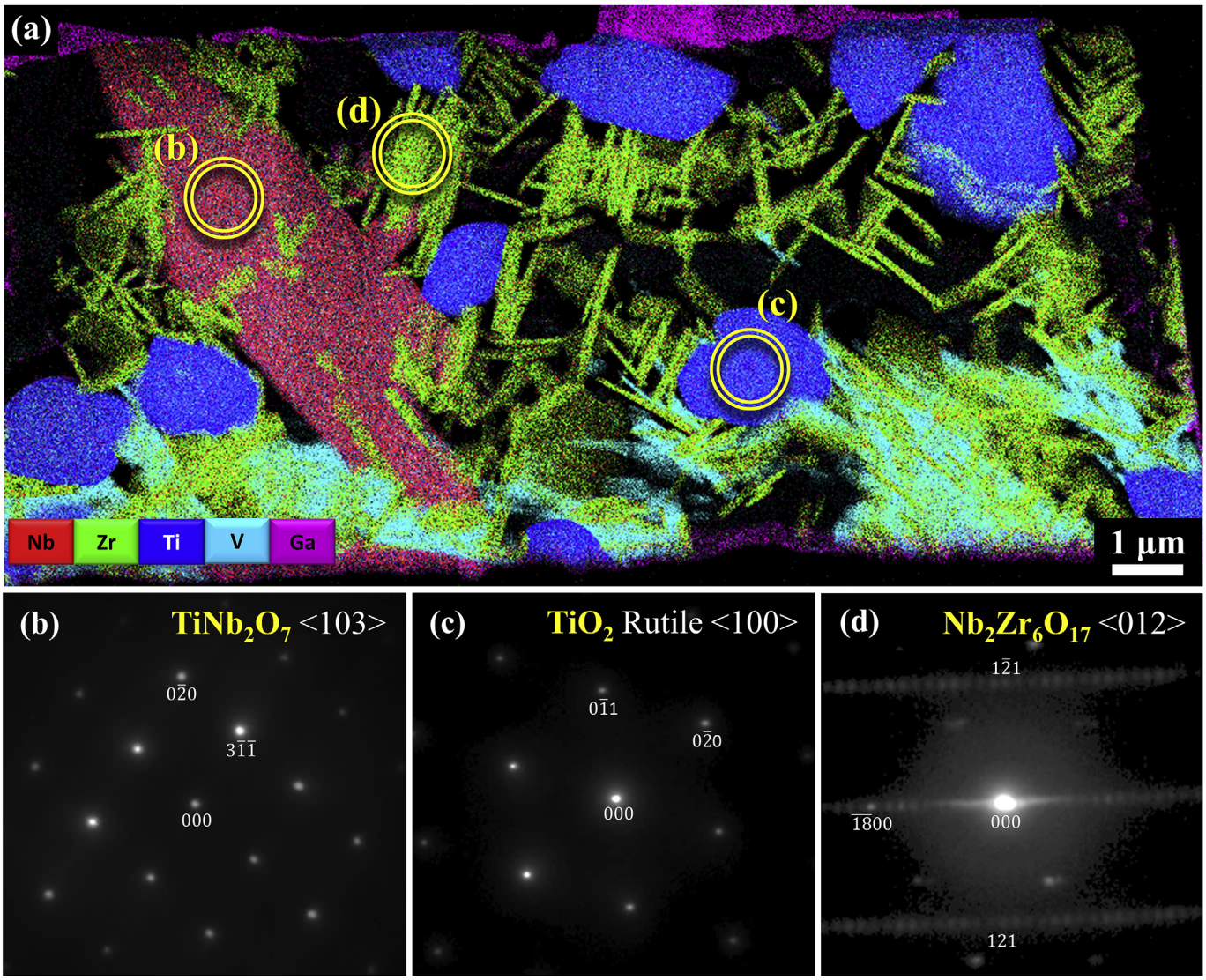


Fig. 8. Combined TEM-EDS maps for the NbTiZrV 100hr oxide sample (a) and representative SADPs from each region of the oxide (b)–(d). The SADPs were captured from regions b, c, and d, as labeled in (a).

Table 4
TEM-EDS data (at. %) from the oxide scale formed on the NbTiZrV alloy after 100 h of oxidation. The relative metallic elements are compared to the matching prototype structures and the proposed oxide is given.

Region	Nb	Ti	V	Zr	Matching Structure	Proposed Oxide
Blue	7.6 ± 0.5	83.6 ± 1.5	5.6 ± 0.8	3.2 ± 0.5	TiO_2	TiO_2 + background
Green	23.4 ± 0.8	4.1 ± 0.1	2.5 ± 1.0	69.9 ± 0.6	$Nb_2Zr_6O_{17}$	$(Nb+V)_2(Zr+Ti)_6O_{17}$
Green + Cyan	14.9 ± 1.2	3.0 ± 0.2	39.6 ± 4.0	42.5 ± 2.7	–	$(Nb+V)_2(Zr+Ti)_6O_{17}$ + V_2O_5 (**XRD**)
Red	62.4 ± 0.7	18.9 ± 0.8	14.2 ± 0.1	4.6 ± 0.1	$TiNb_2O_7$	$(Ti+Zr)(Nb+V)_2O_7$

experimentally determined oxide compositions were found to be in good agreement with the proposed prototype crystal structures.

Fig. 9 shows STEM-HAADF images and high-resolution elemental chemistry maps for the NbTiZrCr oxide scale. The scale was more dense/uniform than that from NbTiZrV (Fig. 7) and consisted of a fine mixture of high-Z and low-Z contrast oxides, Fig. 9(a) and (f). The dark-features are voids. The individual elemental maps showed a preference for Zr (green) to segregate to the high-Z contrast phase, while Nb, Ti, and Cr all shared an affinity for the low-Z contrast phase, Fig. 9(b)–(e). This chemical preference is shown more clearly in the higher magnification STEM-HAADF image and corresponding chemistry map in Fig. 9(f)–(g). Additionally, some regions of Cr-enrichment were observed in two areas, as indicated by the white arrows in the elemental Cr map in Fig. 9(c). It is thought that these particular regions represent remnant former Laves phase regions to some degree.

The stacked elemental maps are shown over a larger sample area in Fig. 10(a). Four distinct regions arise, which include Zr-rich (green) regions, Nb/Ti/Cr regions (purple), Cr-rich regions (cyan) and voids (black). SADPs captured from the purple Nb/Ti/Cr regions (b) matched a structure consistent with $NbCrO_4$ (ICDD 00-034-0366, P42/mnm (No. 136)). Conversely, patterns taken from the

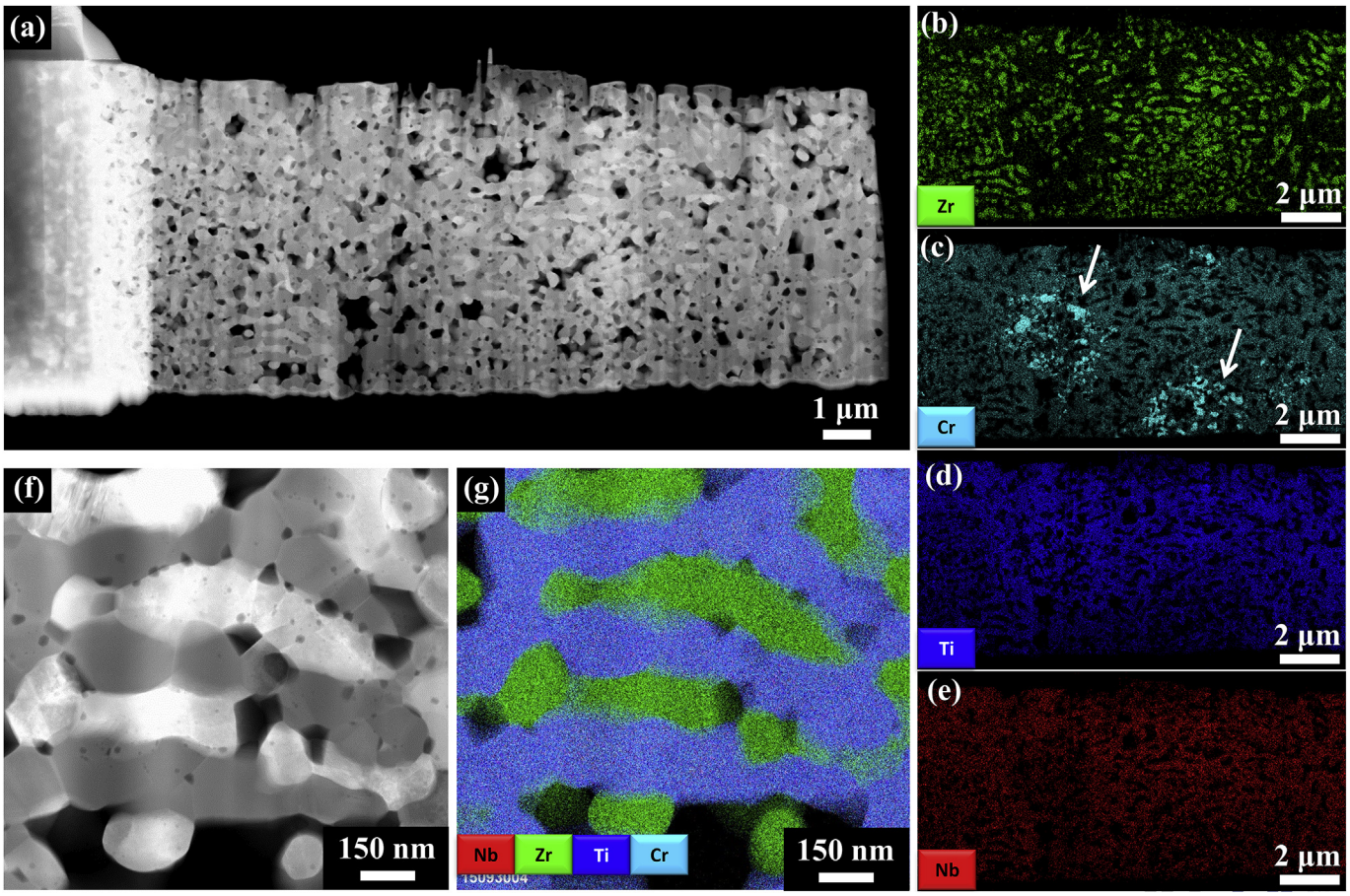


Fig. 9. STEM-HAADF image of a specimen lifted-out of the center of the NbTiZrCr 100hr oxide scale (a) and corresponding TEM-EDS maps (b)–(e). An additional higher magnification STEM-HAADF image taken from (a) and corresponding stacked TEM-EDS map are shown in (f)–(g).

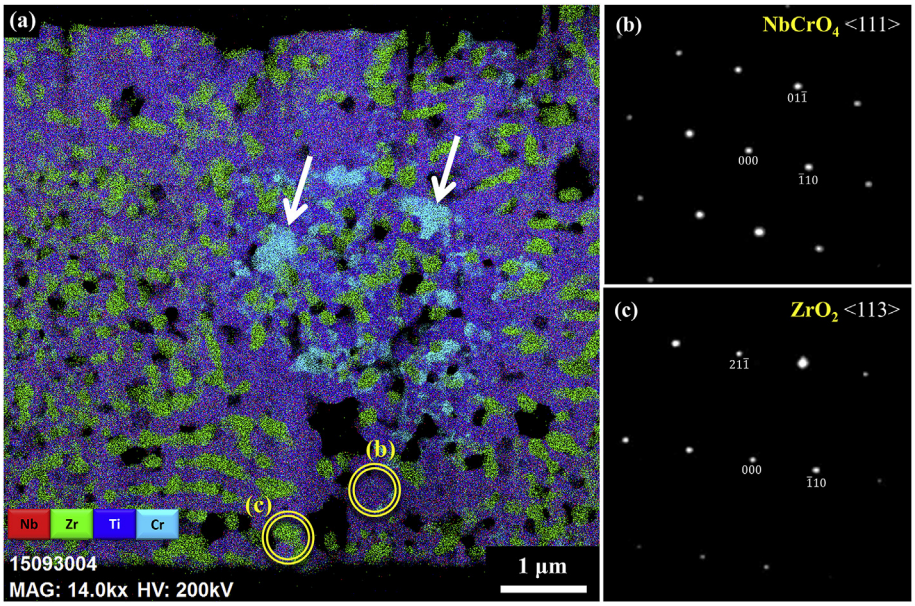


Fig. 10. Combined TEM-EDS maps for the NbTiZrCr 100hr oxide sample (a) and representative SADPs from each region of the oxide (b)–(c). The SADPs were captured from regions b and c as labeled in (a).

green Zr-rich regions (c) matched ZrO_2 (ICDD 00-036-0420, P21/c (No. 14)). Unique patterns could not be obtained from the Cr-rich regions, but it is plausible to suggest that these regions are likely Cr_2O_3, originating from the former Cr-rich Laves phase. As was the

case for the NbTiZrV oxides, the respective lattice parameters for the oxides in the NbTiZrCr scale were also not found to be significantly different in comparison to the prototype structure lattice parameters.

Chemical analysis was conducted using a TEM-EDS detector for the NbTiZrCr oxide phases, and the results are shown in Table 5. The green regions, which matched a ZrO_2 structure, consisted almost entirely of Zr. Thus, it was proposed that this phase was likely ZrO_2. The purple regions, which matched a $NbCrO_4$ structure, were proposed to be some variation of $(Nb+Cr+Ti)O_4$. The actual site preference of this structure could not be identified based on the experimental ratios of the elements. However, it is plausible that some level of substitutional variation of Ti exists on both Nb and Cr sites. As for the cyan Cr-rich regions, the composition was almost entirely Cr, thus Cr_2O_3 is highly likely.

To further validate the proposed oxide structures for both alloys, XRD was conducted in plan-view from 3.60×10^5 s (100 h) oxidized specimens. The peak locations for the suggested prototype structures are overlaid on the experimental spectra in Fig. 11. For the NbTiZrV alloy, the reference peaks match the experimental spectrum quite well, indicating good consistency with TEM results, Fig. 11(a). Additionally, V_2O_5 peaks are shown for comparison, due to the high concentration of "unassociated" V in the TEM chemistry maps. This indicates that there is likely a combination of V_2O_5 and Zr-rich oxide near the bottom of the TEM foil, where trivial SADPs could not be obtained. A similar result was observed for the NbTiZrCr alloy, where the proposed prototype structures again matched well with the experimental spectrum, Fig. 11(b). However, additional peaks were also found to correlate with a combination of Cr_2O_3 and NbO. The presence of Cr-oxides can be explained by the Cr-rich chemical segregation found in the TEM-EDS maps, Fig. 9(c). Furthermore, the XRD data were captured from the free surface of the oxide scales, which may contain other non-stable, transient oxides not necessarily found in the center of the scale (e.g. NbO). However, the proposed structures are in good agreement with the oxide structures determined via TEM analysis.

3.3.2. Abbreviated oxidation tests

Short term oxidation tests were conducted for 1800 s (0.5 h) and 1.44×10^4 s (4 h) in air to better understand initial stages of oxidation and identify preferred sequences of oxide formation. These time periods were selected in order to explore the two distinct regimes of oxidation (with $n = 0.74$ and $n = 0.38$) observed for the NbTiZrCr alloy, Fig. 2(b). It was expected that similar oxide morphologies/mechanisms would be observed at both conditions for the NbTiZrV alloy since a single linear regime of oxidation was sustained throughout long term tests.

Fig. 12 shows digital images of the alloys before oxidation (a), after 1800 s (0.5 h) of oxidation (b), and after 1.44×10^4 s (4 h) of oxidation (c). From the respective color change, it is clear that both alloys formed a relatively thick and adherent oxide scale after a short duration of oxidation, Fig. 12(b). After 1.44×10^4 s (4 h) of oxidation testing, the NbTiZrCr alloy displayed some additional oxide growth, which was accompanied by a slight change in volume and color. However, for the NbTiZrV alloy, the oxide scale was observed to grow rapidly, as evident from the substantial size change, Fig. 12(c). Fig. 12(d) shows a rescaled mass plot containing the 3.60×10^5 s (100 h) TGA data with additional data points representing the specific mass change for the abbreviated oxidation tests. In both cases, the short term tests adhered to similar trends as observed in the long-term TGA oxidation tests.

Fig. 13(a)–(c) shows representative cross-section BSE images of the NbTiZrV alloy after 1.44×10^4 s (4 h) of oxidation. Similar to the long term tests (Fig. 3(b)), the oxide scale was relatively thick (~1550 μm including complete oxidation and partial internal oxidation) and displayed evidence of a small, initial, Nb-rich external scale with Zr-rich needles, Fig. 13(a)–(b). However, the majority of the oxidation appeared to progress internally until reaching a central un-oxidized region labeled "bulk" in Fig. 13(b). The evolution of the oxide scale was highly complex and consisted of complete internal oxidation near the free surface followed by a series of partially oxidized regions moving toward the un-oxidized center of the alloy. The predominantly single phase alloy showed signs of a eutectoid-type reaction during the first stages of internal oxidation, which abruptly transitioned to complete oxidation, as shown by the dashed boundary line in Fig. 13(c). To better understand this initial oxide formation, TEM analysis was conducted on a sample lifted-out of the "first internal oxidation stage", as indicated in Fig. 13(c). Fig. 13(d)–(j) show a representative STEM-HAADF image of this region along with corresponding TEM-EDS elemental maps. This particular region contained both high-Z and low-Z regions, each containing a fine-scale precipitation product, Fig. 13(d). Based on TEM-EDS, the low-Z regions were heavily enriched in Zr (green), while the high-Z regions were rich in Nb (red) and V (cyan). In addition, Ti-rich (blue) precipitates with plate-like morphologies were observed in both regions and were distributed at the phase boundaries in a globular morphology, Fig. 13(e). It is also important to note that the Zr-rich regions also contained Nb/V rich laths as well. Through careful interpretation of the TEM-EDS maps, Fig. 13(f)–(j), it was determined that the regions rich in Zr or Ti were also correspondingly rich in oxygen, which is also consistent with their relative atomic-Z contrast. Conversely, regions rich in Nb and V were lean in oxygen. Based on this chemical information, it is thought that the high-Z and low-Z phase mixture is likely Nb+V-rich BCC and ZrO_2, respectively. In addition, the Ti-rich phase, distributed in all regions, is likely TiO_2. These assumptions could not be verified by electron diffraction due to the relative size and complex distribution of phases/oxides. Additionally, these elements have a high solubility for oxygen and could also be oxygen enriched metal for short oxidation times. A comparison of the 1.44×10^4 s (4 h) microstructure with the 1800 s (0.5 h) condition revealed that a fairly linear, internal oxidation front tended to consume the alloy and steadily progressed with longer oxidation exposure. Mechanistically, this correlates well to the single oxidation regime ($n = 0.91$) observed throughout long term testing when considering specific mass change alone, Fig. 2(b). With that being said, the oxidation mechanism was still observed to proceed via a series of complex steps.

Fig. 14(a)–(c) shows representative cross-section BSE images of the NbTiZrCr alloy after 1.44×10^4 s (4 h) of oxidation. The extent of oxidation was observed to be ~512 μm in depth, with only ~25 μm being complete oxidation and the remaining portion only partially

Table 5
TEM-EDS data (at. %) from the oxide scale formed on the NbTiZrCr alloy after 100 h of oxidation. The relative metallic elements are compared to the matching prototype structures and the proposed oxide is given.

Region	Cr	Nb	Ti	Zr	Matching Structure	Proposed Oxide
Green	0.8 ± 0.8	0.9 ± 0.6	2.3 ± 0.8	95.9 ± 2.0	ZrO_2	ZrO_2
Purple	33.2 ± 0.5	32.4 ± 0.8	30.6 ± 2.2	3.9 ± 1.0	$NbCrO_4$	$(Nb,Cr,Ti)O_4$
Cr-Rich Areas	84.7 ± 3.0	3.6 ± 0.9	8.6 ± 0.7	3.1 ± 3.0	–	Cr_2O_3 (**XRD**)

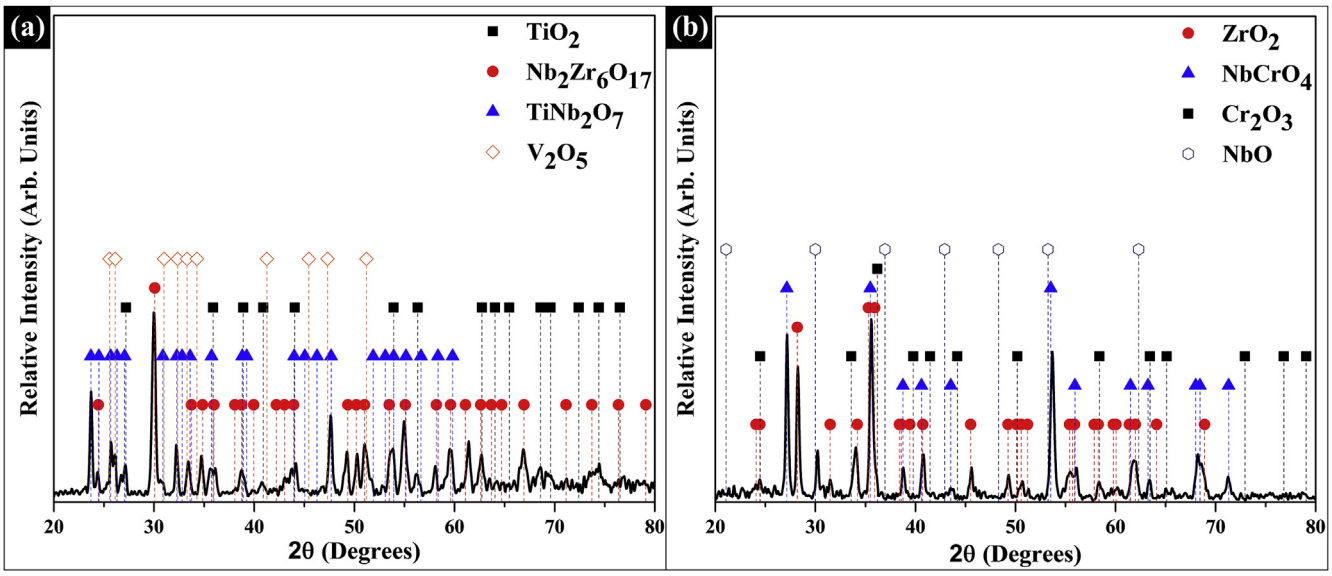


Fig. 11. Plan-view XRD spectra captured from the free surface of the 100hr oxide scales for the NbTiZrV alloy (a) and NbTiZrCr alloy (b). The respective structures identified via TEM are displayed for comparison.

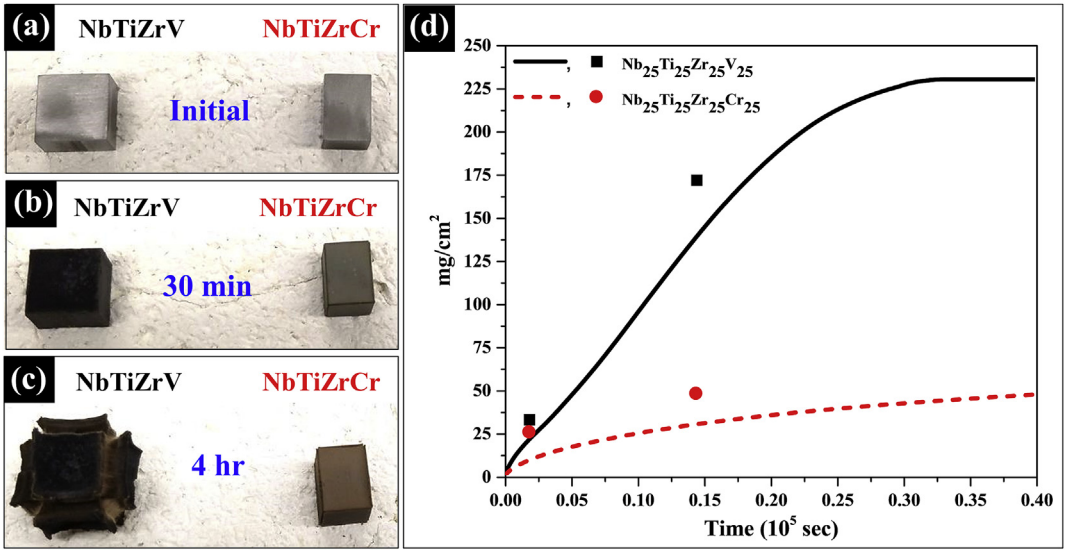


Fig. 12. Representative digital images of the NbTiZrV and NbTiZrCr alloys after abbreviated oxidation for 0 h (a), 30 min (b), and 4 h (c) at 1000 °C; and corresponding abbreviated specific mass change plot showing the 100hr TGA data and two data points from 30 min to 4 h.

oxidized based on the oxygen content, Fig. 14(a)–(b). In contrast to the NbTiZrV alloy, plan-view and cross-sectional imaging of the NbTiZrCr alloy revealed no distinct proclivity for external oxide scale formation, whereas the primary mechanism was observed to be internal oxidation, Fig. 14(a). In the initial stages of internal oxidation (nearest the un-oxidized core of the alloy), the prior BCC phase was found to transform into a lamellar mixture of high-Z and low-Z phases, Fig. 14(c). This morphology is consistent with the results from the longer oxidation tests. The internal oxidation front also easily progressed along the prior BCC phase, but appeared impeded to some degree by the Cr-rich Laves phase. Partial oxidation of the Laves phase induced some internal fine-scale precipitation products, Fig. 14(c).

To better understand the initial oxide products, a TEM foil was lifted-out along a prior-BCC/prior-Laves phase boundary, as shown by the STEM-HAADF image and corresponding TEM-EDS elemental maps in Fig. 14(d)–(j). Similar to the NbTiZrV alloy, Zr and Ti rich phases formed from the former matrix phase due to the presence of oxygen and were also oxygen rich. Nb-rich regions from the former matrix phase and Cr-rich regions from the former Laves phase were lean in oxygen. However, some of the Cr-rich regions in the prior-BCC regions, did exhibit oxygen enrichment. Using the previously defined chemical assumptions with oxygen enrichment, it is probable that the Zr-rich (green) and Ti-rich (blue) phases observed in the former matrix phase formed ZrO_2 and TiO_2 during a shorter period of time, while the Nb-rich (red) phase remained BCC for a longer time. Closer chemical inspection of the former Laves phase revealed a combination of Laves, Cr_2O_3, and fine-scale ZrO_2. Again, these assumptions are based solely on chemical data. Thus, the former BCC phase appears to transform into a mixture of Nb-rich BCC and ZrO_2 lamellae, which are often separated by TiO_2 (in the form of round particles or lamellae) and a small extent of Cr_2O_3/

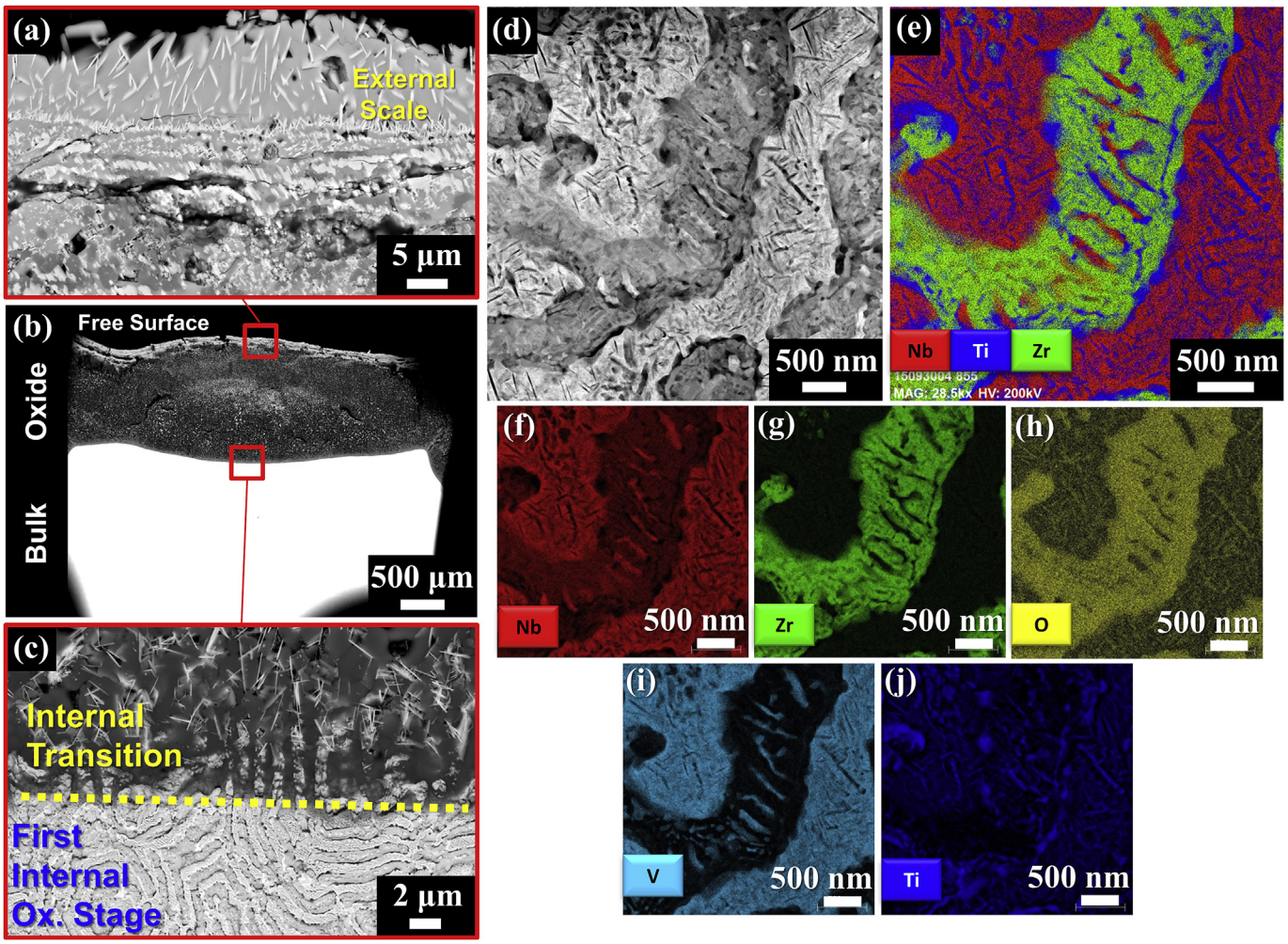


Fig. 13. Cross-section BSE images of the NbTiZrV alloy after 4 h of oxidation at 1000 °C (a)–(c); and STEM-HAADF image with corresponding TEM-EDS maps (d)–(j) captured from a sample lifted out of the first internal oxidation stage shown in (c).

Laves particles. The former large Laves particles appeared to consist of a mixture of fine ZrO_2 and Cr-rich precipitates. Surprisingly, little Ti remained in the former Laves particles, although a continuous layer of Ti+O-rich phase (likely TiO_2) formed continuously around each former Laves particle. Below this TiO_2 layer, the former Laves particles exhibited a Cr-rich layer, but this layer was also correspondingly leaner in oxygen.

The NbTiZrCr alloy samples oxidized at 1.44×10^4 s (4 h) and 1800 s (0.5 h) exhibited similar oxide products. However, one distinct difference was the establishment of a complete oxidation layer/scale in the 1.44×10^4 s (4 h) sample that was not present in the 1800 s (0.5 h) condition. Interestingly, this external oxide scale appears to develop solely due to internal oxidation. This primary difference between the two oxidation conditions could explain the shift in oxidation mechanism from ($n = 0.74$) to ($n = 0.38$) after approximately 1.8 h. Under this logic, the established, external oxide layer provides some level of oxidation resistance.

4. Discussion

The oxidation mechanisms of both the NbTiZrV and NbTiZrCr alloys are highly complicated. Both of the alloys in this work exhibited primarily internal oxidation. This tendency is likely due to the high concentration of reactive elements (e.g. Ti, Zr, Nb), most of which exhibit a high solubility for oxygen at elevated temperatures, combined with the lack of a protective external scale, which subsequently allows for the rapid ingress of oxygen. Additionally, the preferred oxide products formed in this work appear to be governed more so by thermodynamic stability than kinetics. Interestingly enough, it was observed that in these alloys simple oxides tended to form initially during short oxidation times and then in most cases reacted to form more complex oxides. It was determined that these reactions were favorable in reducing the oxidation rates in the NbTiZrCr alloy, but not in the NbTiZrV alloy.

To better comprehend this oxide evolution, the initial oxide structures from the abbreviated oxidation tests were compared with the 100hr oxide products. Fig. 15(a)–(b) shows respective TEM-EDS maps for the NbTiZrV alloy. For simplicity and ease of viewing distinct regions, V is not shown in Fig. 15(a), but it was already demonstrated that Nb and V tended to segregate to similar regions, Fig. 13(f) and (i). Based on the strong similarities between the long term and abbreviated oxide products, probable reaction pathways are proposed. It is thought that the "globular" TiO_2 particles formed at the boundaries of the Nb+V-rich and Zr-rich lamellae in the abbreviated oxidization tests remain throughout and merely coarsen with prolonged time. Comparable regions are identified with arrows in Fig. 15(a)–(b). As for the Nb+V-rich lamella and TiO_2 strips inside them (Blue and Red), it is proposed

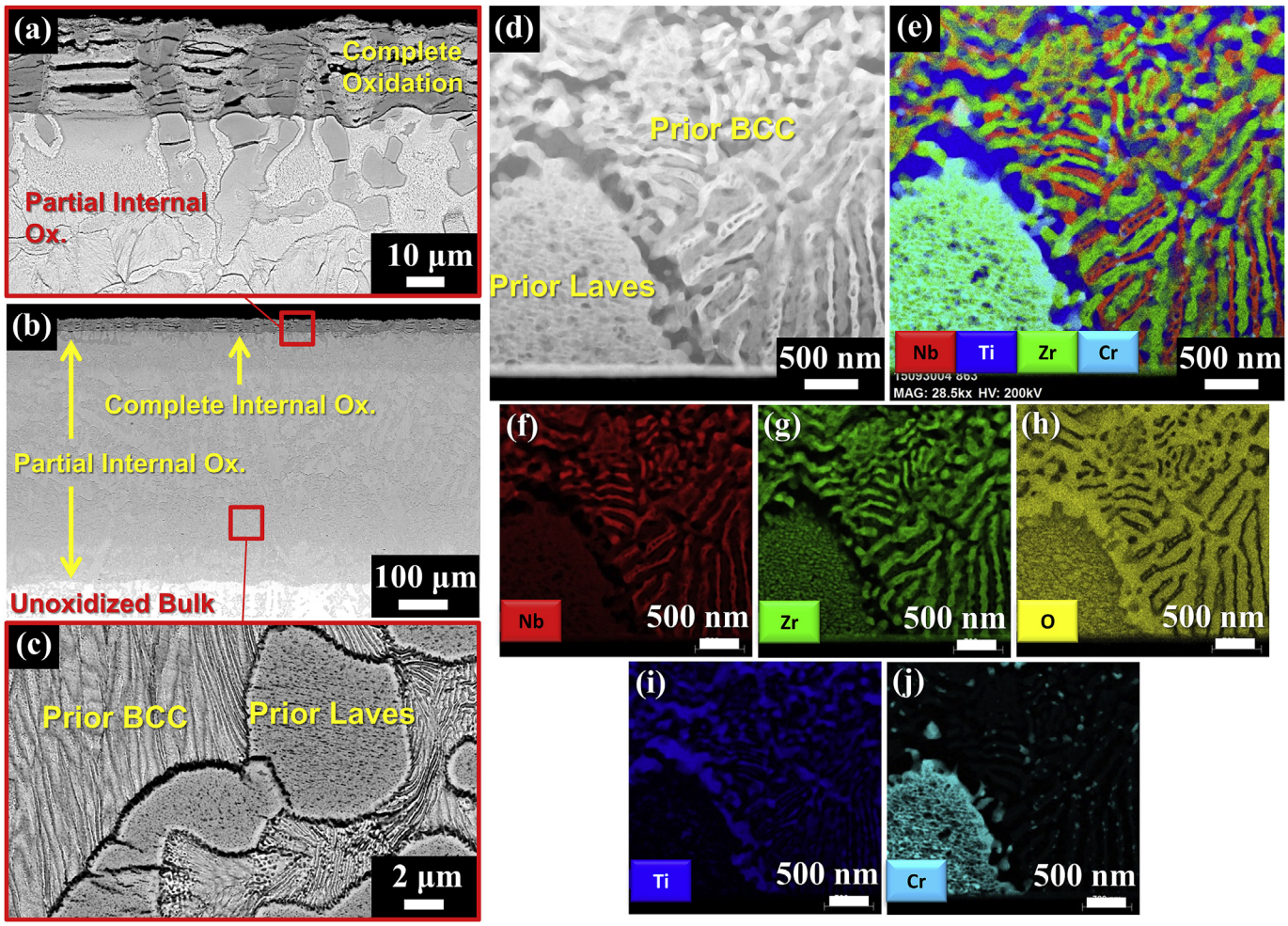


Fig. 14. Cross-section BSE images of the NbTiZrCr alloy after 4 h of oxidation at 1000 °C (a)–(c); and STEM-HAADF image with corresponding TEM-EDS maps (d)–(j) captured from a sample lifted out of the region shown in (c).

that a subsequent reaction of Nb with TiO_2 and oxygen promotes the formation of $TiNb_2O_7$. This specific reaction is noted under Fig. 15(b) and is consistent with prior studies on the high temperature oxidation behaviors of binary Nb-Ti alloys [33,34]. A similar process is thought to occur for the ZrO_2 and Nb+V-rich BCC regions (Green/Red). Here there is a large concentration of ZrO_2, which can ultimately react with Nb once it oxidizes to form Nb_2O_5 [32]. This reaction path could ultimately explain the strong presence of $Nb_2Zr_6O_{17}$ in the 100 h scale. The oxide structure is also structurally anisotropic (Ima2: a = 40.92 Å, b = 4.93 Å, c = 5.27 Å) being elongated along one primary axis, which may explain the drastic change to a rod-like morphology, Fig. 15(b) [32]. In these reactions, V likely separates from Nb and forms its own oxides, some of which are volatile (V_2O_5, melting point 690 °C) or can be dissolved in water (V_2O_3, V_2O_5). It is likely that during holding at 1000 °C and/or the sample preparation for metallography, these oxides evaporate or dissolve leaving voids. This may explain why the V_2O_5 is recognized in a noticeable amount by XRD and hardly seen in SEM samples.

A similar comparison was made for the NbTiZrCr alloy, as shown by the TEM-EDS maps in Fig. 16(a)–(b). It was previously determined that the prior-BCC and prior-Laves regions both formed ZrO_2. In contrast to the NbTiZrV alloy, it is thought that this ZrO_2 ultimately remains after 100 h of oxidation, Fig. 16(b). It was then determined that the long term oxide scale also contained $NbCrO_4$, with equal amounts of Nb, Cr, and Ti, along with some regions of Cr_2O_3. Unfortunately, based on the remaining mixture of phases determined for abbreviated oxidation tests (e.g. TiO_2, Nb-rich BCC, Laves, and Cr_2O_3), the transformation path to the $NbCrO_4$ structure remains inconclusive. No distinct parallels considering the oxidation behavior of concentrated NbTi or NbCr alloys could be determined. In addition, the authors have no knowledge of oxidation work that has been conducted on concentrated NbTiCr ternary alloys. With that being said, future work investigating the oxidation behavior of the concentrated binary and ternary alloys is currently underway in order to better understand the sequence of reactions governing the behavior of the NbTiZrCr alloy and their temperature dependence.

Due to the strong predominance of internal oxidation, thermodynamic parallels for oxide formation can be made for both alloys. Figs. 15(c) and 16(c) show standard free energy of formation (per mole) for the relevant oxides [35]. In all cases, the two most thermodynamically stable oxides (lowest standard free energy of formation) were TiO_2 and ZrO_2, which were found to internally oxidize first and drive subsequent oxidation reactions with remaining metal and/or later oxides to form complex structures. This ultimately led to sluggish oxidation kinetics for the NbTiZrCr alloy. Thus, it may be possible to tailor composition to promote specific phases/distributions enriched in desired elements to promote a particular sequence of oxide reactions. This concept would

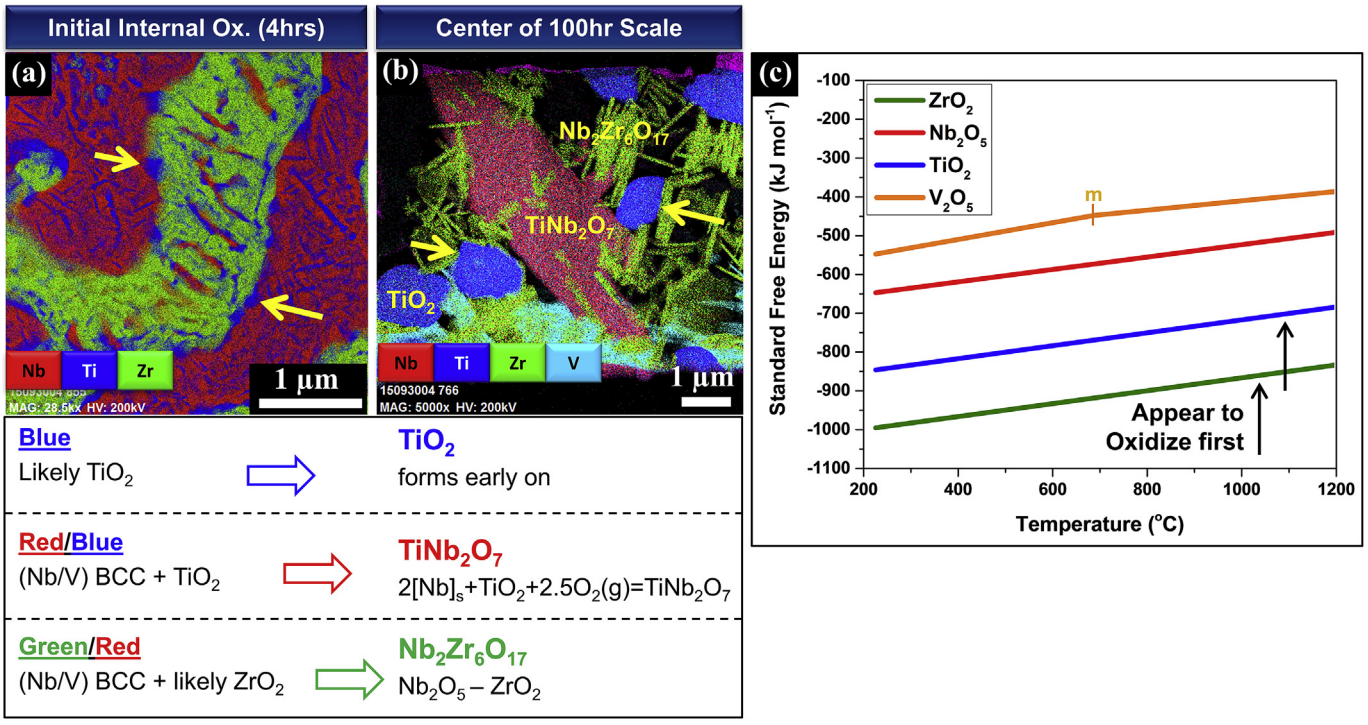


Fig. 15. Stacked TEM-EDS maps from the oxides formed on the NbTiZrV alloy after 4 h (a) and 100 h (b) at 1000 °C, along with proposed reaction paths. A plot of standard free energy of formation for relevant oxides is shown in (c).

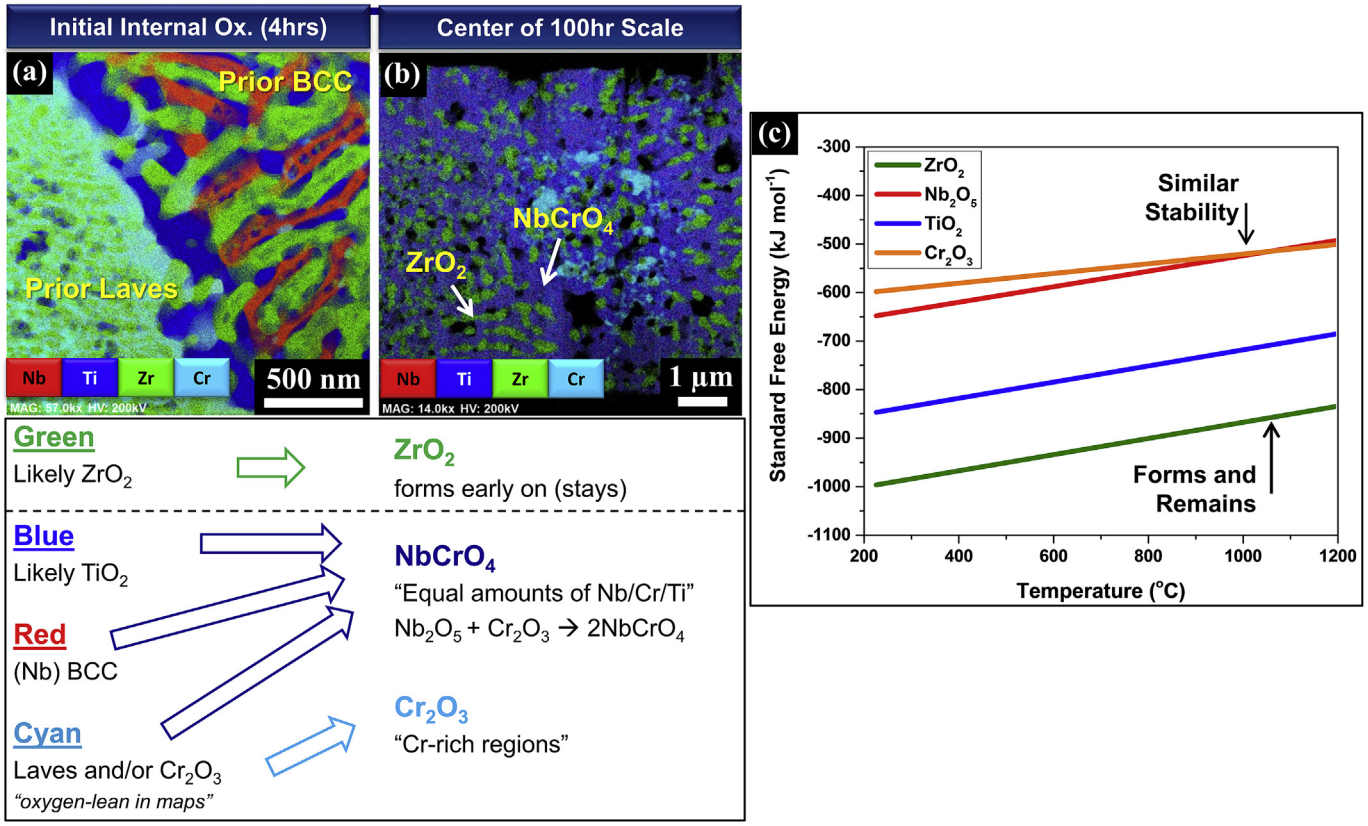


Fig. 16. Stacked TEM-EDS maps from the oxides formed on the NbTiZrCr alloy after 4 h (a) and 100 h (b) at 1000 °C, along with proposed reaction paths. A plot of standard free energy of formation for relevant oxides is shown in (c).

ultimately require prior knowledge of the thermodynamic tendencies and an understanding of the oxidation behaviors of the relevant binary/ternary systems involved. Additionally, thermodynamic data for many of the complex oxides observed in this work are not readily available.

To this point, the most interesting aspect of this work is that the sluggish oxidation kinetics observed for the NbTiZrCr alloy did not rely on the presence of classically protective, external oxide layers such as Al_2O_3, Cr_2O_3, and SiO_2. Instead, the oxide was found to contain a mixture of $NbCrO_4$ and ZrO_2. Since it is well known that ZrO_2 does not provide adequate resistance to the ingress of oxygen and was dispersed all throughout the oxide scale, it is assumed that the primary source of the sluggish oxidation was due to $NbCrO_4$. This particular oxide, $NbCrO_4$, has a rutile-type structure and is similar to spinel type oxides that one might observe during the initial oxidation of M-Cr-Al type alloys (e.g. $NiCr_2O_4$, $NiAl_2O_4$). It is thought that the formation of $NbCrO_4$ tends to neutralize the presence of Nb_2O_5 by the reaction of Nb_2O_5 with Cr_2O_3. This alternative concept of promoting oxidation resistance in refractory alloys is consistent with one effort from the 1960s that proposed the use of complex oxides like $NbCrO_4$ for diffusion barriers to protect Nb-Cr alloys [36]. Interestingly enough, this same oxide was a primary constituent in the scale of the NbTiZrCr alloy.

One of the goals of this effort was also to directly compare the effect of Cr and V on the oxidation behavior of these two alloys. While they do show a difference in performance it is hard to elucidate what aspect is responsible for the change. As stated above the V containing alloy shows predominantly a single phase and displays a slightly more predictable evolution of oxide formation as the ingress of oxygen occurs. Since the presence of Cr leads to the precipitation of a secondary phase, the progression of internal oxidation is more complex. Secondary phases potentially allow for multiple oxides forming at the same time due to the different chemical potential of each phase, the presence of phase boundaries which can influence oxygen mobility, and no one phase that is a direct parallel in composition to the solid solution present in the V alloy. It should be noted that it is also hard to discern quantitatively at this point how much of the improved oxidation behavior is due to the chemical or microstructural difference, since both alloys formed non-continuous, multi-phase, oxide scales. It is also well known that microstructure can play a significant role in oxidation and in this particular case requires further investigation.

Inherently, the oxidation kinetics are unlikely to approach those observed for alloys that form classically protective oxide scales (e.g. Al_2O_3, Cr_2O_3, SiO_2), yet the authors propose the promotion of complex oxides as an alternative, but realistic approach to promote a significant improvement in the oxidation resistance of RCCAs and potentially dilute refractory alloys. It is envisioned that this alternative concept could allow for transition of these alloys for short, aggressive environments in uncoated form or may provide some level of risk mitigation for coated alloys in the case of coating compromise. Future oxidation work in this compositional space is still required and may reveal other complex oxides that also offer some degree of oxidation protection. While the alloys in this work primarily displayed internal oxidation, it is proposed that if one could facilitate the formation of external scales with these chemical species or promote the earlier formation of sluggish oxides, an even greater level of oxidation resistance could be achieved.

5. Conclusions

This work investigated the fundamental oxidation behaviors of NbTiZrV and NbTiZrCr refractory complex concentrated alloys (RCCAs) at 1000 °C in air. The following conclusions were made:

1) The oxidation of the NbTiZrV alloy was rapid and exhibited nearly linear behavior with time ($n = 0.91$) out to 3.60×10^5 s (100 h) at 1000 °C in air. The entire alloy sample transformed to oxide after only ~ 8 h of oxidation time. Investigation of the long term scale revealed four discernable oxides with the following prototype structures: $TiNb_2O_7$, TiO_2, $Nb_2Zr_6O_{17}$ and V_2O_5. This oxide mixture did not appear to aid in oxidation resistance. The oxidation proceeded primarily via internal oxidation.
2) The oxidation kinetics for the NbTiZrCr alloy were nearly linear ($n = 0.74$) out to 6581s (1.8 h) and then transitioned to nearly parabolic ($n = 0.38$) out to 3.60×10^5 s (100 h) at 1000 °C in air. This was primarily attributed to the formation of a complete external oxidation layer. Investigation of the long term scale revealed two discernable oxides with the following prototype structures: $NbCrO_4$ (with nearly equal amounts of Nb, Cr, and Ti) and ZrO_2. The oxidation proceeded primarily due to internal oxidation.
3) Abbreviated oxidation tests showed that the preference for internal oxidation led to the sequential oxidation of the most stable oxide formers (i.e. Ti to form TiO_2 and Zr to form ZrO_2) in both cases. These oxides then tended to react with either remaining metal or other oxides to form internal scales consisting of a mixture of complex and simple oxides (e.g. TiO_2, $TiNb_2O_7$, and $Nb_2Zr_6O_{17}$ for the NbTiZrV alloy). The specific sequence of reactions for the NbTiZrCr oxide mixture is still not well understood.
4) Some degree of improvement to the oxidation resistance of RCCAs can be attained using complex oxides.

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